

ST CROIX ROHLSSEN, Virgin Islands, U.S.

WMO: 785510

Lat: 17.700N

Lon: 64.813W

Elev: 19

StdP: 101.10

Time Zone: -4.00 (W04)

Period: 00-19

WBAN: 11624

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.0	20.8	15.7	11.2	24.1	16.6	11.8	24.3	10.3	27.2	9.6	27.4	2.4	0	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	32.1	25.8	31.7	25.7	31.3	25.6	27.0	30.7	26.7	30.4	26.4	30.2	7.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.2	29.0	25.5	20.8	28.9	25.2	20.4	28.8	84.6	30.7	83.2	30.5	82.0	30.2	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	9.1	8.4	DB	17.5	34.1	3.0	3.3	15.3	36.4	13.5	38.4	11.8	40.3	9.6	42.7	
			WB	16.7	27.7	1.0	0.8	16.0	28.3	15.4	28.8	14.8	29.3	14.1	29.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	25.2	25.5	25.7	26.4	27.1	28.2	28.6	28.6	28.2	27.9	26.8	26.1	(6)
		DBStd	1.58	1.12	1.06	1.08	1.00	1.09	0.97	0.83	0.83	1.08	1.17	1.14	1.13	
		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
		CDD10.0	6220	473	434	487	493	531	547	575	577	547	554	504	499	
		CDD18.3	3178	214	201	228	243	273	297	317	318	297	296	254	241	
		CDH23.3	31271	1445	1439	1749	2163	2732	3397	3687	3762	3416	3217	2342	1923	
(14)	Wind	WSAvg	4.7	4.6	4.9	4.7	4.7	4.9	5.4	5.6	5.0	3.9	3.9	3.9	4.6	(14)
		PrecAvg	1073	57	48	48	65	101	68	80	105	135	133	136	98	
		PrecMax	2004	187	91	105	217	454	199	176	306	543	412	415	327	
		PrecMin	690	14	5	10	0	6	3	12	36	27	27	28	20	
		PrecStd	276	34	25	25	53	101	49	44	66	99	88	99	71	
		DB	29.7	30.2	30.2	30.7	31.2	32.1	33.0	33.0	33.3	32.2	31.4	30.7		
		MCWB	23.3	23.1	23.8	24.5	25.2	25.6	25.8	26.2	26.6	25.9	25.4	24.5		
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	29.2	29.6	29.7	30.2	30.7	31.5	32.0	32.0	31.9	31.6	30.8	30.0		(20)
		MCWB	23.4	23.2	23.7	24.2	25.0	25.5	25.6	25.9	26.0	25.7	25.2	24.3		
		DB	28.8	29.1	29.2	29.7	30.2	31.1	31.4	31.6	31.3	31.1	30.2	29.4		
		MCWB	23.3	23.1	23.4	24.0	24.9	25.4	25.5	25.8	25.9	25.6	25.1	24.1		
		DB	28.3	28.6	28.8	29.2	29.7	30.7	31.0	31.1	30.9	30.6	29.7	28.9		
		MCWB	23.1	22.9	23.2	23.8	24.8	25.3	25.4	25.7	25.8	25.6	24.9	23.9		
		WB	25.1	25.2	25.5	26.1	26.5	27.0	27.1	27.2	27.4	27.1	26.6	25.6		
(28)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	27.8	27.8	28.3	29.3	29.8	30.4	30.7	31.0	32.7	31.3	29.8	28.8		(28)
		WB	24.4	24.2	24.7	25.3	25.9	26.5	26.6	26.7	26.8	26.6	26.0	25.1		
		MCDB	27.5	27.7	28.0	28.8	29.4	30.1	30.4	30.6	30.7	30.4	29.5	28.6		
		WB	23.9	23.7	24.1	24.8	25.5	26.1	26.2	26.5	26.5	26.2	25.6	24.6		
		MCDB	27.2	27.5	27.7	28.4	29.0	29.9	30.1	30.3	30.4	30.1	29.0	28.1		
		WB	23.5	23.3	23.7	24.3	25.1	25.7	25.8	26.1	26.2	25.9	25.2	24.2		
		MCDB	26.8	27.1	27.3	28.0	28.6	29.5	29.8	30.0	30.1	29.8	28.5	27.6		
(35)	Mean Daily Temperature Range	MDBR	6.1	6.2	6.2	5.9	5.4	5.3	5.5	5.6	5.9	5.9	5.9	5.8		(35)
		5% DB	6.4	6.6	6.3	6.2	5.9	5.5	5.8	6.0	6.3	6.2	6.3	6.2		
		MCDBR	2.8	2.8	2.7	2.5	2.4	2.2	2.3	2.4	2.6	2.6	2.7	2.7		
		5% WB	6.1	6.2	5.8	5.6	5.1	5.3	5.5	5.6	6.0	5.8	5.7	5.6		
		MCWBR	3.0	2.9	2.8	2.6	2.5	2.4	2.5	2.6	2.8	2.7	2.8	2.8		
		taub	0.369	0.375	0.380	0.405	0.448	0.513	0.517	0.477	0.437	0.392	0.378	0.374		
		taud	2.583	2.561	2.544	2.497	2.387	2.165	2.149	2.300	2.442	2.587	2.620	2.606		
(42)	Clear-Sky Solar Irradiance	Ebn at Noon	918	929	931	902	852	791	789	828	865	901	904	903		(42)
		Edh at Noon	94	101	106	111	123	152	154	134	115	97	90	90		
		RadAvg	4.94	5.75	6.34	6.63	6.40	6.44	6.55	6.45	6.03	5.34	4.73	4.58		
(44)	All-Sky Solar Radiation	RadStd	0.12	0.11	0.22	0.34	0.39	0.36	0.29	0.29	0.37	0.36	0.30	0.20		(44)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Regional (7 neighbors)	+0.23	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	+79	+80	

Nomenclature: See separate page